

HAEMOSPORIDIAN INFECTION DYNAMICS AND IMMUNE GENE VARIATION IN A
POPULATION OF HYBRIDIZING CHICKADEES

by

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ABSTRACT

Parasites are a complex and diverse group of organisms that have long intrigued researchers due to their spectrum of host specialization. Similarly, the consequences that parasites can impose on hosts also exhibit variation ranging from inconsequential to severe, with direct effects on fitness. Host-parasite interactions that are highly consequential can play a significant role in shaping host species evolutionary histories. Infectious outbreaks, and the lifespan of the organisms they directly impact, are a flicker along the timescale of evolutionary history, but the effects of these events can live on in genomes for millennia. One tool that researchers can use to investigate questions about evolutionary pattern and process are natural hybrid zones, regions where species overlap in their range and reproduce offspring of mixed ancestry. The focus of my dissertation is on the potential role of parasites in shaping host species barriers within natural hybrid zones and on understanding how this can translate to host genome evolution. I first conduct a review of the literature where I introduce a novel framework for thinking about parasites and how they might shape host species barriers in animal host systems. My third and fourth chapters are both an examination of haemosporidian parasites (*Plasmodium*, *Leucocytozoon*, and *Haemoproteus*) in a sympatric population of two songbird species – the black-capped (*Poecile atricapillus*) and mountain (*Poecile gambeli*) chickadee. First, I report on an invasive *Plasmodium* parasite that is infecting this population of

chickadees, with some thoughts on the ecology and evolution underlying parasite range expansions. I then systematically characterize the haemosporidian communities infecting this chickadee population and compare their divergent infection dynamics. These infection dynamics can influence species barriers and subsequently host genome evolution. I therefore focus my fifth chapter on understanding the evolution of the songbird genus, *Poecile*, with a focus on their immune genes. The findings of this work suggests both historic and ongoing introgression amongst *Poecile* species, with potential selection on immune genes in two *Poecile* hybrid zones. Through this dissertation I aimed to harness the investigative power of natural hybrid zones to explore host evolution as it is shaped by parasitism.

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DEDICATION

For Theo

PREVIEW

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PREVIEW

CHAPTER 1

INTRODUCTION

Understanding the genomic mechanisms underlying the generation and maintenance of species barriers (i.e., reproductive isolation between taxa) has been a shared goal of evolutionary biologists since the modern synthesis (a combined framework for evolutionary biology and genetics; Huxley 1942). This dissertation investigates species barriers and the potential role that parasites play in shaping them, both historically and in the present time. The reader should know that species barriers are always in flux, they are context dependent and complex, sometimes frustrating to delineate, and, as with many phenomena in biology, our definition of what they are is often too simplified and thus easy to misconstrue. Species barriers do not have a singular mechanism that shapes them.

Species barriers are defined as the mechanisms that prevent genetic exchange between populations that are different enough to be considered species, though the definition of species is itself complex. These mechanisms can act either before mating or fertilization (i.e., prezygotic) or after fertilization (i.e., postzygotic) (Mayr 1963; Coyne and Orr 2004). Pre-zygotic barriers can include physical distance that causes populations to be isolated from one another preventing their interbreeding. For example, populations of lava lizards on the Galápagos Archipelago (*Microlophus albemarlensis* complex) are genetically distinct between islands (Jordan and Snell 2008), which is due largely to the effects of genetic drift, where isolated populations stochastically lose genetic variation and population structure develops over time. Pre-zygotic barriers can also occur without

physical distance, in which case they are sometimes mediated by sexual selection. For example, Turbek et al. (2021) found that two incipient species of capuchino seedeater (*Sporophila iberaensis* and *S. hypoxantha*) have markedly similar genomes, but exhibit assortative mating that is mediated by male song and plumage color. Post-zygotic barriers prevent genomic introgression following interbreeding and occur within the hybrid offspring produced by divergent parental species (Coyne and Orr 2004). For example, Sandstedt et al. (2021) examined hybrids from experimentally crossed species of monkeyflower (*Mimulus* spp.) and reported seed inviability and male and female sterility. While the idea of species barriers is essential for testing questions in evolutionary biology, it is important to remember that species barriers can erode, that they are often dependent on environmental context, and that their strength can vary over evolutionary time (Barton and Hewitt 1989; Martin and Jiggins 2017; Taylor and Larson 2019).

The breakdown of species barriers means that both pre- and post-zygotic isolation is overcome, and that different species interbreed and produce hybrid offspring that can in turn backcross with parental species (Harrison 1993). This process can lead to introgression—the movement of genomic regions between species—which can occur differentially along genomes. The analysis of the resulting patterns may lend insight into genes, and gene combinations, that are important for local adaptation and potentially exhibit selection (Harrison and Larson 2016). For example, analysis of introgression patterns at multiple genes of hybridizing saltmarsh (*Ammodramus caudacutus*) and Nelson's (*A. nelsoni*) sparrows suggested selection for tidal marsh adaptations between these taxa (Walsh et al. 2016). Mechanisms, like adaptation to a salt marsh, that might support or prevent introgression are varied and change over time. Therefore, we cannot define these mechanisms in a way that is applicable across all hybrid zones (regions where

distinct species come in contact and hybridize at varying rates). While it is impossible to define the entire set of these mechanisms, and the extent of their individual roles in shaping species barriers, testing the role of a given mechanism can provide us with insight on the processes of evolution and speciation. For this dissertation, I investigated the potential role of parasites in shaping patterns of, and the process that leads to, introgression for animal host species.

Parasites are ubiquitous, they are some of the most diverse taxa on this planet, and they can have a marked impact on selection exhibited by their hosts (Coltman et al. 1999; Poulin and Morand 2000). For instance, parasite-mediated selection was found to be a powerful driver preventing inbreeding amongst an island population soay sheep (*Ovis aries*) in St. Kilda Scotland (Coltman et al. 1999). Inbred sheep were more susceptible to infections with intestinal parasites that were linked to higher mortalities. Host-parasite systems have also played a prominent role in shaping our understanding of evolution (Anderson and May 1979; Hamilton and Zuk 1982; Altizer et al. 2003). For example, the Hamilton-Zuk hypothesis postulated that parasites influence male sexual signals and subsequently influence female mate-choice and male fitness (Hamilton and Zuk 1982). The extent to which parasites contribute to host speciation, and potentially adaptive radiations, has also been considered in previous literature (Karvonen and Seehausen 2012; Brunner and Eizaguirre 2016; El Nagar and MacColl 2016). However, there are few empirical studies of parasites and speciation, and a significant knowledge gap surrounding our understanding of how parasite-mediated selection can influence patterns of introgression between hybridizing host taxa. My dissertation addresses this knowledge gap by examining parasitism in host hybrid zones using a combination of field studies, molecular methods, and recent advances in genomic tools.

In Chapter 2, I review our current understanding of how parasites might play a role in shaping species barriers in animal hybrid zones (Theodosopoulos et al. 2019). Sage et al. (1986) first reported that within a hybrid zone of two house mouse species (*Mus musculus musculus* and *M. m. domesticus*) hybrid mice appear more susceptible to infections with parasitic worms. Since then, researchers have considered the potential divergence of infection dynamics within versus outside of hybrid zones. Multiple studies have taken this approach and focused on plant-herbivore interactions (McClure 1985; Fritz et al. 1998; Horgan and Crisol 2013) with a notable review of some earlier work by Fritz et al. (1999). When I conducted the literature search for Chapter 2, I decided to direct my efforts on animal hybrid zones in attempts to keep my research focused on parasitism, and to develop a framework that can relate parasites directly to their effects on animal host species barriers. Of additional importance is the advent of next generation sequencing technology that allows researchers to accurately identify levels of genomic introgression in taxa sampled from natural hybrid zones (Gompert et al. 2017). These recent advances allow researchers to identify hybrids more accurately and explicitly ask questions about parasitism across a continuum of introgression in natural study systems.

Parasites, like their hosts, are complex and exhibit their own selective pressures where speciation is sometimes imparted directly from host-parasite interactions. For example, a study by Villa et al. (2019) demonstrated rapid reproductive isolation in feather feeding lice from the same natural population that was experimentally isolated and evolved on birds with divergent feather sizes across ~60 generations. While species of feather-feeding lice often exhibit tight-coevolution with their hosts (Clayton and Johnson 2003) some parasites are highly generalist and can infect multiple host species

that are markedly diverged (Cooper et al. 2012; Martínez-de la Puente et al. 2021). In Chapter 3 I report on the range expansion of the SGS1 strain of the avian malaria parasite, *Plasmodium relictum* (Theodosopoulos et al. 2021). *Plasmodium relictum* is one of the most widespread parasites among avian taxa (Martínez-de la Puente et al. 2021). Range expansions of parasites like *P. relictum* can lead to severe population declines and even extinction events for the affected novel hosts (Hatcher et al. 2012). For example, the introduction of the invasive fly, *Philornis downsi* to the Galápagos Archipelago has led to severe declines for some of Darwin's finches (Fessler and Tebbich 2002; Fessler et al. 2010; Koop et al. 2011). Considering the ranges of parasites, their ecological drivers, and infection dynamics with hosts, is important for understanding how parasites can modulate a given host study population. In the same vein, host-parasite infection dynamics likely play a role in the potential for parasites to shape host species barriers.

Host evolutionary histories shape host-parasite interactions of the present. Previous studies have compared parasite infections across host taxa (Brooks 1988; Poulin and Morand 2000; Esser et al. 2016; Clark et al. 2018) but few have explicitly investigated these dynamics between closely related host species. Chapter 4 of this dissertation is focused on the parasite communities and infection dynamics (i.e., proportion of infected individuals within the sample population, infection intensity, and potential influence of infection on body condition) within a population of two closely related chickadee species that overlap in sympatry—the black-capped (*Poecile atricapillus*) and mountain (*Poecile gambeli*) chickadee. While these species occasionally hybridize (Grava et al. 2012) previous research suggests they diverged from one another in the early Pleistocene (Gill et al. 1993). Given their likely historic isolation we hypothesized that black-capped and mountain chickadees, while closely related and infected with similar parasite

communities, would exhibit divergent infection dynamics. To test this hypothesis, we examined their haemosporidian parasites. Haemosporidians are a broad group of organisms, including the genera *Plasmodium*, *Leucocytozoon*, and *Haemoproteus*, that infect the blood cells of their vertebrate hosts during some portion of their life cycle (Valkiunas, G. 2004). If infection dynamics are markedly different between host species, then I consider how this in turn might influence genomic exchange within this population of hybridizing chickadees.

Introgression following species barrier breakdown does not occur equivalently across the genome (Harrison and Larson 2016). In considering parasites as mediators of species barriers, and their effects on genomic exchange, I explore immune gene variation in Chapter 5 across a clade of birds in the genus *Poecile*. This chapter broadly characterizes phylogenetic clustering of taxa based on randomly sampled genes across the genome in comparison to a diverse assembly of immune genes. I further investigate patterns of introgression within this clade in addition to genomic exchange and potentially associated patterns of selection in two avian hybrid zones— the black-capped and mountain chickadee hybrid zone, and the black-capped and Carolina (*P. carolinensis*) chickadee hybrid zone. Previous research on the potential role of parasites shaping species barriers within these hybrid zones has been conducted (See Chapter 4, Rice et al. 2021) and this chapter builds on the prior literature by exploring genetic exchange involving immune genes, an essential component of host-parasite infection dynamics.

Lastly, Chapter 6 is a synthesis of my dissertation research, where I revisit the question— To what extent are parasites able to play a role in shaping species barriers? I then discuss the contributions resulting from each chapter of this work.

CHAPTER 2¹

PARASITES AND HOST SPECIES BARRIERS IN ANIMAL HYBRID ZONES¹

2.1 ABSTRACT

Species barriers are tested in hybrid zones when gene flow occurs between hybridizing species. Hybridization can erode species barriers, lead to the introgression of adaptive traits, or remain stable through time. Outcomes in hybrid zones are influenced by divergence between the hybridizing taxa, behavior, ecology, and geography. Parasites and pathogens play a major role in host fitness and appear to have varied impacts on species barriers in hybrid zones. We comprehensively reviewed the literature on parasitism in animal hybrid zones and present an evolutionary framework within which to consider parasite-hybrid interactions. Parasites most frequently show potential to contribute to species barrier breakdown in hybrid zones, but also frequently show potential to facilitate the maintenance of species barriers. Incorporating eco-immunology, parasite community theory, and spatiotemporal approaches will be important as genomic tools allow researchers to examine parasites and hybrid zones at greater resolution and in a diversity of natural habitats.

2.2 PARASITES AS MEDIATORS OF SPECIES BARRIERS

Understanding the mechanisms that underlie the generation, maintenance, and erosion of species barriers is a primary goal of evolutionary biology (Coyne and Orr 2004). Nonetheless, identifying these mechanisms and their consequences has posed a

¹ This chapter was published in *Trends in Ecology and Evolution* with Scott A. Taylor and Amanda K. Hund

consistent challenge for evolutionary biologists (Mayr 1942; Coyne and Orr 2004). To gain a better understanding of the speciation process, researchers have regularly used hybrid zones (Harrison and Larson 2014; Kenney and Sweigart 2016; Mořkovský et al. 2018), regions where interbreeding occurs between individuals from two populations or groups that differ in one or more heritable characters (Harrison and Larson 2014). Hybrid zones are often viewed as naturally occurring laboratories for understanding the maintenance or erosion of reproductive isolation: they are regions where divergent genomes are mixed, testing new allelic combinations and the permanence of species barriers (Barton and Hewitt 1985). Furthermore, hybrid zones are ubiquitous; they occur across a broad range of landscapes, in a multitude of ecosystems, and between a taxonomically diverse set of species.

When hybridization occurs, species barriers will either erode due to introgression (Toews et al. 2016), or be maintained by low hybrid fitness, which, when extreme, manifests as hybrid inviability (Mack and Nachman 2017). Low hybrid fitness can be the result of many factors including (but not limited to) Bateson-Dobzhansky-Mueller incompatibilities (BDMI) (Dobzhansky 1936; Muller 1942; Orr 1995), hybrid sterility (Lee et al. 2008), competition, and behavioral isolation (Hatfield and Schluter 1999), and maladaptation to parental environments (Hatfield and Schluter 1999). One overlooked mechanism that might be important in shaping the relative fitness of hybrids is the impact of parasitic infections (Mouliá 1999). Parasites, given their ubiquity in nature, their host-specificity, the tight co-evolutionary relationships they have with their hosts, and their critical influence on host fitness (Laanto et al. 2017), might be playing an underappreciated role in shaping species barriers (Karvonen and Seehausen 2012).

Throughout, we use the term parasite to refer to both macro and micro-parasites (i.e., pathogens).

Evidence suggests that parasites can influence species barriers in a diversity of ways. Parasites frequently reduce hybrid survival, but hybrids sometimes appear better equipped to resist parasites compared to parental species, or have the same impact on hybrids as on parental taxa (Mouliat 1999; Table S2.1). Genomic admixture in hybrid organisms can lead to variation in, among other things, their immunological function (e.g. recombination of immune gene alleles) (Racimo et al. 2015) and ecological interactions (e.g. habitat use, resource competition) (Tanaka 2007). Such variation can have a marked impact on a host's response to parasitism. Therefore, parasites can have a significant, though varied, influence on the evolutionary trajectory of their host species within hybrid zones.

Although the impact of parasitism in hybrid organisms has been the focus of numerous studies, early investigations were generally lab-based or focused on economically important taxa (Bakke et al. 1999; Mouliat 1999). More recently, studies of animal hybridization and hybrid zones have proliferated (Table S2.1). Previous reviews on the topic used a framework developed by Fritz et al. (1994) to understand patterns of hybrid resistance mechanisms (Fritz et al. 1994; Strauss 1994; Fritz 1999; Mouliat 1999). These reviews were generally focused on understanding immunological changes in hybrids, and how patterns differ between plant and animal hybrids, but not on understanding the consequences of parasitism in hybrid zones for species barriers and evolution (Fritz et al. 1994). Our review shifts the focus from comparing immunological changes in hybrids of different taxa, to understanding the consequences of parasitism for the maintenance or erosion of species barriers, via gene flow, in animal systems. Several exciting advances

have occurred over the past two decades that will facilitate this shift and allow us to examine parasites in hybrid zones in greater detail than ever before.

From significant advances in sequencing methods and computational tools we can now leverage genome-spanning data to identify both parasites and hybrids in nature, and measure gene flow. At the same time, it has become increasingly evident that immunology is fundamentally shaped by ecological interactions in wild populations, which our ability to document and monitor has also increased. With our increasing understanding of eco-immunology (Table 2.1), parasite community ecology (Table 2.2), and the spatiotemporal variation that exists within populations (Table 2.3), we can begin to think more holistically about the impact that parasites are having on species barriers.

Table 2.1. Ecological Immunology in Hybrid Zones: In animals, the immune system is the last and most important line of defense against parasite infections. Variation in immunity can thus have important consequences for host fitness and survival and the transmission of parasites between individuals. The field of ecological immunology has worked to explain individual variation in immunity by understanding costs and trade-offs between immune function and both resources and other life history traits (Brock et al. 2014). It also seeks to understand both the genetic and environmental contributions to heterogeneity in immune function (Sheldon and Verhulst 1996; Norris and Evans 2000; Schulenburg et al. 2009). By applying concepts and tools from ecological immunology to the study of hybrid zones, we can gain a better understanding of the underlying mechanisms shaping host-parasite interactions, and how these interactions influence species barriers.

There are several predictions for how hybridization might lead to variation in immune investment, and the type and magnitude of immune function that individuals mount (Fritz 1999; Moullia 1999). Variation in immunity can be generated by genetic differences and admixture, or by phenotypic plasticity and the environment. Teasing apart these differences is important for understanding the evolutionary consequences of immune heterogeneity and how this can impact species barriers. To effectively understand these mechanisms, it is important to recognize that the immune system cannot be characterized as a single static trait (Ahmed and Hashish 2006; Zylberberg 2015). Instead, it should be measured with a multifaceted approach and with the recognition that it is a dynamic and complex system operating in a certain ecological context (Hedrick 2017). We suggest that combining several different immune measures within a hybrid zone will give a richer view of how hybrid immune function might differ from parental species. In addition, it is important to recognize that hosts have other behavioral, physical, and chemical defenses against parasites that might also differ in hybrids. This table summarizes links between hybrid immune function, mechanisms, and implications for parasite infections and species barriers.

Hybrid Immune Function Relative to Parental Taxa	Potential Mechanisms	Implication for Parasite Infections	Implications for Species Barriers
Increased	-Increased MHC diversity -Admixture of locally adapted resistance alleles	-Increased resistance -Fewer parasites -Reduced transmission	-Erode species barriers via increased gene flow: hybrids have immune advantage
Intermediate	-Combination of parental immune alleles -Intermediate MHC diversity -Intermediate use of environment	-Intermediate levels of parasites -Greater diversity of parasites	-Adaptive introgression of resistant immune alleles.
Same as parental taxa	-Little genetic difference in immunity between parental taxa -Immune response environmentally driven	-Similar parasite infections	-Parasites do not influence species barriers
Decreased	-Hybrids have increased stress that lowers immune functions -Hybrids have fewer resources to invest in immunity -Metabolic and immune dysfunction due to genetic mixing	-Decreased resistance -More parasites -Increased transmission	-Reinforce species barriers via reduced gene flow: hybrids have an immune disadvantage.

Table 2.2. Tackling the complexity of parasite communities

Within hybrid zones, research has largely focused on hosts and single species of parasites, yet in nature, hosts are infected with a community of different parasites (Bordes and Morand 2011; Vaumourin et al. 2015). Coinfections can exhibit pairwise and higher-order interactions with direct links to the host and indirect connections between parasites via host condition and immunity (Telfer et al. 2010; Johnson and Hoverman 2012). These interactions can lead to diverse outcomes that are not additive and thus cannot be predicted from studying single parasites alone (Rynkiewicz et al. 2015). Host state is thus an emergent property of the host + parasite community, which is often dynamic and spatially complex (Penczykowski et al. 2016).

Tools and recent advances in community ecology and network theory can be adapted to gain new insights into these interactions (Johnson et al. 2015; Walker et al. 2017). These tools offer a bridge from fine scale interactions within individuals to population-level patterns of coinfection and susceptibility in different host types (i.e., hybrids and parentals). For example, trait-based approaches help to identify functional characteristics that shape assemblages and interactions, and have been used to understand variation in parasite transmission (Paull et al. 2012; Daniel et al. 2013). Such tools might allow for the identification of functional traits associated with parasites, such as resistance alleles, that are likely to be introgressed in a hybrid zone or change fitness outcomes of infected individuals and thus influence species barriers. Metacommunity approaches, which have recently been applied to disease ecology (Suzán et al. 2015; Johnson et al. 2016; Mihaljevic et al. 2018), offer tools to link host-parasite interactions across local and regional scales, and can be used to partition the role of species interactions and local ecology in shaping parasite distributions within hybrid zones. Such tools might be particularly useful in hybrid zones that cover a large geographic area, or are moving through time, and could help to explain variation in host-parasite interactions within the same study organism (Figure 2.2).

In addition, interaction networks can be used to understand the direct and indirect connections between parasites within an individual host, as well as between hosts (Lafferty 2010; Telfer et al. 2010). Networks assembled from parasite co-occurrence data can be compared for hybrid and parentals to measure variation in susceptibility and parasite community structure (Poulin 2010; Godfrey 2013; Bashey and Bashey 2015). Analysis of network structure can also be used to better understand and quantify variation in parasite communities, which could subsequently be linked to variation in hybridization rate and survival of hybrids ((Dallas and Presley 2014; Dallas and Poisot 2018), Table 2.3). If we are to move beyond a one-parasite framework when studying hybrid zones, we will need to adapt and apply new tools that allow for a complex view of multiple interacting parasites.